Franklin Crypto Research Competition — UBC 2026
Track: **Short, market cap > $1B**

Short PUMP: I Ran Bots Inside Token Factories

Asset: PUMP (pump.fun)
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Terms. Launchpad: a site where anyone mints a token on a bonding curve, a formula that prices the token as people buy and sell. Graduation: the token moves to the PumpSwap exchange. Fees: what traders pay. Revenue: what pump.fun keeps. Take rate: revenue ÷ fees. Run-rate: the last 30 days annualised (the paper also shows 90 days and a full year). Factory: an operator that mints pools by the hundred and drains them. Family: pools sharing one contract bytecode. Drain: the operator pulls the liquidity or sells into the buyer. Address: one wallet. Strict P&L: unsold tokens counted at zero. Second buyer: a buy from outside the creator's wallets within 60 seconds. Perpetual: a futures contract with no expiry. Funding: its periodic payment between longs and shorts. Open interest: the total outstanding. FDV: price × all tokens, circulating or not. Catastrophe stop: the exit if the PUMP/SOL ratio doubles.

Disclosure. From September 23 to October 13, 2026 I ran small automated trading bots on launchpad pools on four chains, with $56–122 of outside capital (Appendix C). I hold no position in PUMP or SOL.

AI use: an assistant helped me draft and translate the text from my notes in French, and helped with the numbers, the sources, the Allium queries (on definitions I fixed beforehand), the scripts and the figures. The collector, the bots, the thesis and the choices are mine.

Main report: 5 pages (executive summary = page 1). Title page, references and appendices are outside the page count. Figures 1–5 are in the main text; Table B1 and Figures A1–D3 are in the appendices.

Executive summary

In one line. PUMP is priced for a peak that is already behind us. I ran bots with real money inside token factories and came out ahead, then pointed the same tools at pump.fun: its crowd loses every day and has started to leave, and the buyback that holds the price stops in April. Short PUMP against SOL, target $0.0022.

What I did. From September 23 to October 13 I ran small buying bots from 53 addresses on four chains, with about a hundred dollars of outside capital, and made **+$682 to +$748** (Appendix C). The operators adapted at their own pace: my edge lasted three weeks on Arc and four days on Robinhood Chain. The money bought me proof that my replay matches real fills (page 2). I then read every trade on pump.fun's curve for sixteen days with the same tools.

PUMP — SHORT · Price $0.00474 (Oct 21, 2026, ref. 2) · **Target $0.0022, –54 %: today's 2.6× multiple on pre-boom revenue ($398M a year, Feb–Jun) = $1.03B market cap** · Market cap $2.19B = 2.6× run-rate revenue · 50 % of revenue buys back PUMP for one year, changeable by one address · Trade: short PUMP perpetual / long SOL, same amount, no leverage, 1 % of the book · Horizon: April 2027 (key test: Nov 16 weekly print) · What would prove me wrong: weekly revenue ≥ $16M for 4 weeks, buyers > 155k/day (Allium, 7-day mean) for 2 weeks, or launchpad fees ≥ $58M/wk · Catastrophe stop: PUMP/SOL ×2 on a daily close

#1. The crowd loses, and it is leaving. For sixteen days in October my collector recorded every trade on pump.fun's bonding curve for SOL-quoted tokens, 1.69 million positions. The crowd that buys twenty to sixty seconds after a launch lost a median **17.4 %** per position, while a round trip costs 2.5 % in fees. Allium, an independent second pipeline, finds −13 % for the same crowd and days, still five times the fee. Every entry time up to five minutes loses, and so does every one of the sixteen days. It is no October accident: Allium finds the same crowd losing in February, May and August too, a median of 15 %, 12 % and 14 % (Appendix A). And the crowd is leaving: on Allium's count, active buyers fell **16 %** in one week, on all seven same-weekday pairs (Fig. 1), and four weeks after their late-September peak they were 36 % lower, back near their February–June level (Fig. A1).

#2. Priced for one peak week. Before the summer boom pump.fun earned about **$7.7M a week** (February to June). The peak week reached **$17.9M** and the last full week **$14.0M**, the second weekly decline in a row. After both earlier peaks revenue fell, by **87 %** in six months and **52 %** in nine (Fig. 4). At today's multiple of 2.6 times revenue, the pre-boom base is worth **$0.0022** a token. Since my October 21 snapshot, weekly revenue has fallen to $9.7M (week to November 8, DefiLlama) and PUMP has lost about a quarter of its price.

#3. The only programmed buyer has an end date. Half of revenue goes to buying back and burning PUMP, under a one-year policy that runs from April 28, 2026 to about April 28, 2027 (refs 6, 10). Renewal is not promised, and one address can change the share with no delay and no vote (ref. 7). Since the token launched in July 2025, buybacks have taken **$490.7M** of PUMP, more than a fifth of the market cap. PUMP pays no dividend, so this is the only way pump.fun's revenue reaches the token (ref. 20). Against it, **6.875B tokens a month** unlock to the team and investors, who paid little or nothing for them (ref. 8). Over the last month these unlocks were already larger than the buybacks, a net yield of **−1.7 %** a year before the policy even ends.

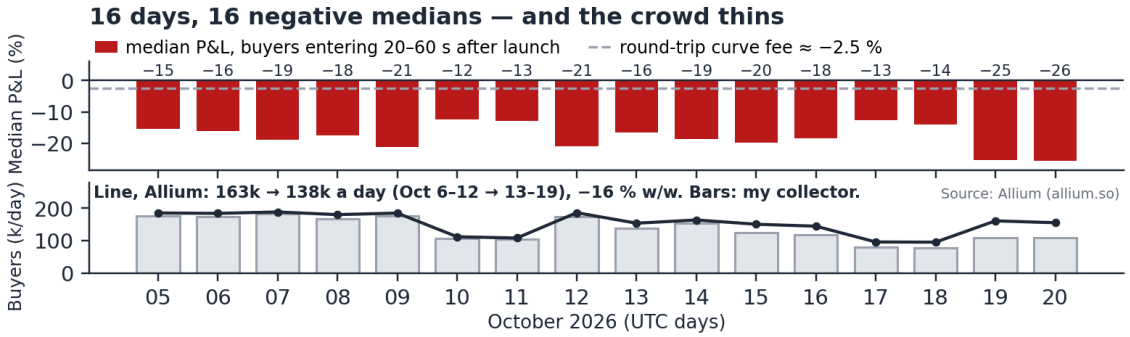


Fig. 1 — The crowd's median position lost money on each of the sixteen days. Top: median P&L of the crowd's positions. Bottom: active buyers per day, my collector (bars) and Allium (line). My collector undercounts after October 14, so weekly buyer counts are Allium's. The two pipelines agree on the direction of every measure, not on most levels (Appendix A). Sources: my collector; Allium query q1 (ref. 19).

Project overview (1): how I worked

I measured before I acted. My collector first showed that most new pools on these chains came from a few factories, and only then did I build the bots. I wrote my rules down before testing them: my method note of September 11, in the repository, set the rule and its thresholds before the test, and the October rules are dated to the minute. The replays had to earn my trust too. One factory family looked like +198 % at the last logged price, but at the real on-chain state of that moment it was −18.9 % (Appendix A).

The factories were no secret. Most new pools on those chains were built to be drained. What I wanted was an indicator that groups pools by factory and spots them early, and I found one: pools that share one contract bytecode belong to one operator. The recipe was to buy $0.10–0.20 of a pool 20 to 45 seconds after it is born, not earlier so as not to hold too large a share of it, and to sell 110 to 240 seconds later, before the drain. I ran it on four chains and made **+$682 to +$748** on the campaign (Appendix C). The replay matched the money: $53.5 a day replayed against $53.1 a day earned, over about five hundred round trips.

The operators fought back (Fig. 2). On Robinhood Chain, with nothing changed on my side, the share of my buys drained before I could sell went from none of 134 on October 4 to 22 % the next day and 10 of 16 on October 6. That evening the whole family stopped launching at 18:44 UTC, and it came back three days later. The same night, none of 166 buys by other traders in pools I did not touch was drained. That is my control. On Arc the rule was a stake wall. Buys of a quarter dollar or more were drained within seconds 156 times out of 157, while my ten-cent buys almost never were. A week later on Arc, pools of the family I was buying died 65 seconds after my buy, against 213 seconds without me. These families behave like small businesses with written rules. One rule I had written in advance missed its own threshold, so I never deployed it.

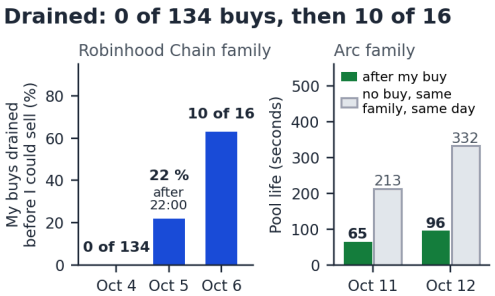


Fig. 2 — How the operators answered my bots: the share of my buys drained before I could sell (one Robinhood Chain family, three days in a row), and how long a pool lived after my buy compared with pools I left alone (one Arc family). All wallets were swept to one public vault at the end (Appendix C). The day-by-day picture is Fig. C1.

This says nothing yet about PUMP's price. Two things carry over to pump.fun. The first is the method, which held with real money. The second is a result: the recipe that won on Arc, replayed on pump.fun's exact curve, wins nothing. All nine of its variants lose on each of the sixteen days, with medians between −2.7 % and −7.3 %. On pump.fun there is no operator for me to outsmart, just a fee on every trade that the crowd pays. The tools also pass a known-answer check: a token that graduates must multiply 14.7 times on the curve, and it does.

Project overview (2): pump.fun under my instrument

The business. pump.fun makes money from four products: the bonding curve, PumpSwap (where graduated tokens trade), the Terminal and the mobile app, and since mid-October perpetuals routed to Hyperliquid. PumpSwap now takes more fees than the curve ($106M in September against $46M), but liquidity providers and creators keep part of them. My own crowd numbers cover the SOL-quoted curve. The company behind pump.fun, Baton Corporation Ltd, is registered in England (directors Noah Tweedale, Alon Cohen, Dylan Kerler). It raised $1.32B in July 2025 at $0.004 a token, and its annual accounts were overdue at the time of the Blockworks filing (ref. 7).

Sixteen days of every trade (October 5–20, 2026). 395,253 SOL-quoted tokens were launched, about a thousand an hour. Half of them found a second buyer within a minute, a count that includes scripted buyers. Yet the median position lost 13.3 % across all entry times, and **17.4 %** for the crowd that enters 20 to 60 seconds after launch. The loss holds on all sixteen days, however it is counted (strict P&L, or −15 % if unsold tokens are valued at the last curve price, Appendix A). The clearest scripted pattern is a creator who sells within seconds of an outside buy about twice as often as at comparable moments (×2.1 after controls, Appendix A). The same scanner read four other launchpads (Table B1, Appendix B).

The crowd is thinning. On Allium's count, active buying addresses went from 163,000 a day in the week of October 6 to 138,000 the next week, while SOL itself lost only 8 %. My collector missed a growing share of the flow after October 14, so here I use Allium's count (Appendix A). Nine months of Allium data put the

peak behind us: 192,000 buyers a day in the week of September 28, and 122,000 four weeks later (Fig. A1). PumpSwap, the part I do not measure myself, is 30 % below its peak week in fees, and its buyers fell by a fifth from September to late October.

The food chain, hand by hand (Fig. 3). I grouped the 1.13 million buying addresses by how many positions each took in the sixteen days. The fish, the 45 % who bought only once, lost a median **46 %**. The 395 busiest addresses, mostly bots with more than two thousand positions each, lost **4 %**, a little more than the fee. Nobody wins on the median. Small fish, with five positions or fewer, are 84 % of the addresses but put in only 11 % of the SOL. The whales, the 1.4 % with more than a hundred positions, put in **62 %**.

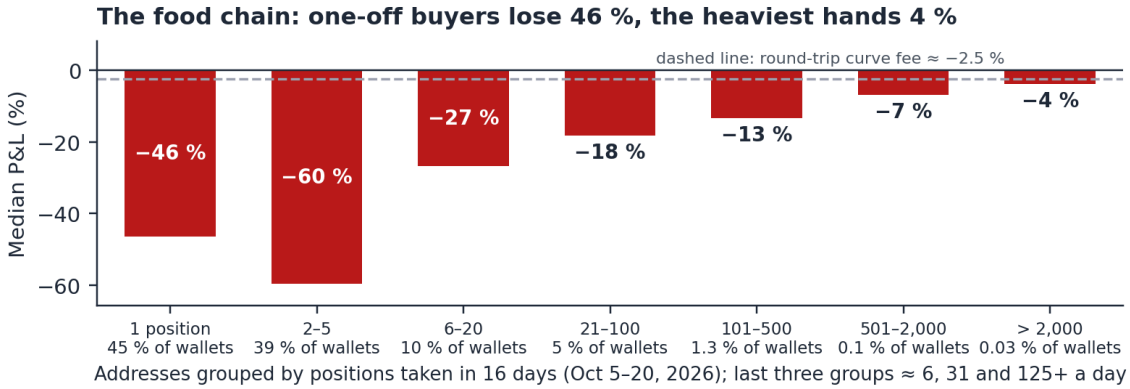


Fig. 3 — The food chain: median P&L per position, by the number of positions an address took in the sixteen days. Source: my Solana collector (second week undercounted, Appendix A), 9.68M positions on non-graduated tokens, strict P&L.

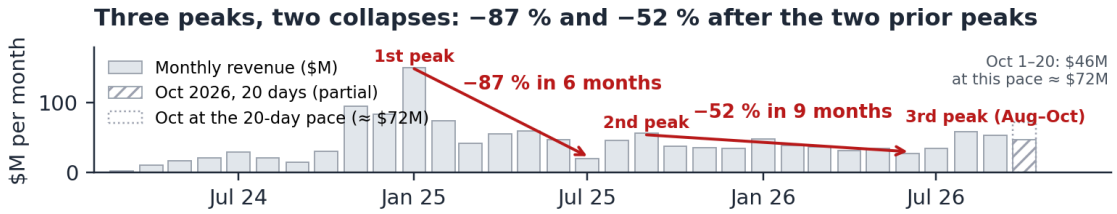
Two public facts (Figs. D1, D3). First, pump.fun now keeps less of each dollar. Fees grew 2.5 times this summer but revenue only doubled, because its share fell. Part of that was a decision (creator rewards, Project Ascend) and part was mix, as PumpSwap grew. Second, the pie shrank (Fig. D3). Across the launchpads alone, weekly fees went from $6.5M in June to $85M in early September, and back to $26M by mid-October. pump.fun kept at least half of launchpad revenue the whole time (ref. 20) and 70–94 % of Solana launchpad swaps (ref. 19), though Pons led on fees for a few weeks around the peak.

Investment proposal (1): what it is worth

Comparables (market cap ÷ last 30 days of revenue, annualised, October 21). PUMP **2.6×** (between 1.6× and 4.2× over the past year) · PONS 3.2× · RAY 10× · JUP 18.3× · HYPE 22×. PUMP is the cheapest of the group on revenue. On the cash it returns to holders, it trades at 5.2 times its yearly buybacks, against about 22 for HYPE. The market already doubts that this revenue will last, and I think it is right.

Target, two methods. *A multiple on base revenue.* Before the boom (February to June 2026), all of pump.fun's revenue (curve, PumpSwap, Terminal and app, ref. 1) averaged $398M a year. At today's 2.6×, that is a market cap of $1.03B, or **$0.0022 a token (−54 %)**. I see 2.6× as generous, because the multiple moves with revenue: it was 1.9× at the June low. On 90 days of revenue the market pays 3.2×, which would give $0.0028, the top of my range. I take the low end, because at base revenue the unlocks exceed the buybacks, and the buyback policy ends in April. *A buyback-coverage price*, a stress test with no multiple. It is the price at which half of revenue absorbs a year of unlocks: $0.0051 at the peak run-rate, within 7 % of today, and **$0.0024** at base revenue. Both methods move in line with revenue, so their agreement proves little on its own.

Revenue base ↓ / multiple →	1.9× (June 2026)	2.6× (today, 30-day basis)	3.2× (today on a 90-day basis; PONS today)
June-2026 trough, $319M/yr	$0.0013	$0.0018	$0.0022
Feb–Jun 2026 average, $398M/yr	$0.0016	**$0.0022**	$0.0028
Trailing 12 months, $465M/yr	$0.0019	$0.0026	$0.0032
Last full week, $728M/yr	$0.0030	$0.0041	$0.0050
Peak run-rate, $836M/yr	$0.0034	$0.0047 (today)	$0.0058



Source: DefiLlama dailyRevenue, pump.fun parent protocol (launchpad, PumpSwap, Terminal, app), Mar 2024 – Oct 20, 2026; October partial (hatched).

Fig. 4 — Three peaks, two collapses: pump.fun monthly revenue since March 2024 (October partial). The weekly series peaked in the last week of September and is –22 % since. Source: DefiLlama (ref. 1), series as of October 21, 2026.

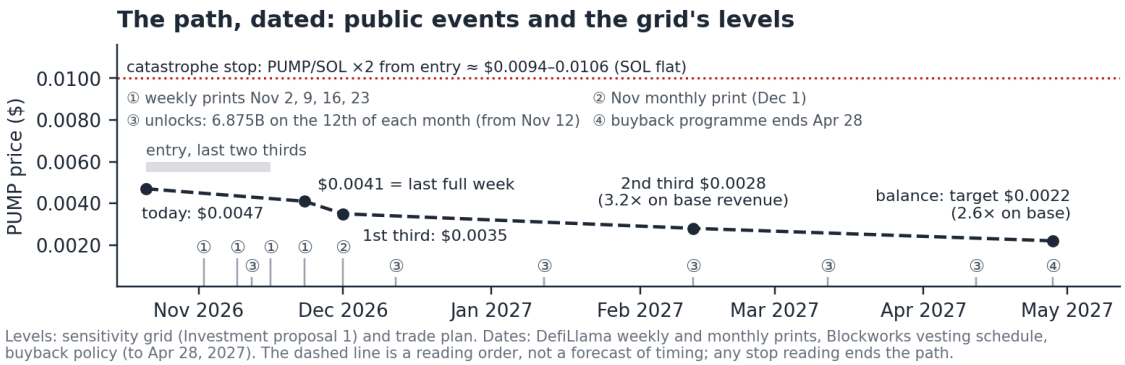
Scenarios (six months). Bear, 25 %: revenue back to its June low at the June multiple, and no renewal of the buyback, $0.0013. Base, 50 %: $0.0022. Bull, 25 %: the boom settles near $15M a week and the market pays 3.25×, $0.0055. Weighted by these odds, the price is about $0.0028. Appendix D runs the reverse check on today's price.

Net supply, and who holds it. PUMP pays no dividend and gives no vote on the buyback (refs 7, 20). The buyback is worth 19 % of the market cap a year at the run-rate and 9 % at base revenue. It is the bulls' best argument, and it has an end date. Each month 6.875B tokens (about $32.6M) unlock to the team and investors (refs 7–8). Insiders kept 92 % of the July cliff (ref. 9), so I treat unlocks as steady supply pressure and use their dates as checkpoints. Over the last month $29.5M was bought back and $32.6M unlocked, a net buyback yield (ref. 14) of −1.7 % a year if every unlocked token is counted. Ownership is concentrated. The ten largest addresses hold 59 % of all PUMP, two thirds of it in accounts that Solscan labels as pump.fun's, and the 430,000 holders outside the top thousand hold 1.3 % between them (refs 4, 19). Most of the rest sits in multisig vaults and large unlabelled wallets that hold rather than sell, so the real float is about a third (ref. 4).

Investment proposal (2): risks and trade

Tested and rejected (or not settled). *The crowd always lost, and pump.fun survived two collapses.* True. My claim is that this crowd is thinning while the price assumes it stays. *The crowd is rotating back to pump.fun.* Its share of launchpad revenue and swaps did rebound (refs 19–20), but the pie is far smaller than at the peak and buyers are 36 % below theirs. That is one of my stop readings. *Fewer but richer buyers.* On Allium's count, revenue per active buyer was flat in the week (+1 %) but still about 60 % above its February–June level, while the number of buyers is almost back to its own. So my target needs each buyer to spend less. That spend sat at $8–10 a buyer-day from February to July and rose only with the boom. The one earlier bump, in May ($12), was gone within four weeks (DefiLlama ÷ Allium, refs 1, 19). *The volume is wash trading.* About 1.3 % of pump.fun volume, on the strict lower bound (ref. 15). I claim the opposite: the volume is real and it loses money, which is worse for how long the revenue lasts.

Catalysts, dated (Fig. 5). The weekly prints of November 2, 9, 16 and 23: a week near $14M is worth $0.0041 at a constant multiple, and I take the first third of the profit below that, at $0.0035. The November monthly print on December 1, the first full month after the peak. The monthly unlock on the 12th, while revenue drifts toward its base ($0.0028 at the higher multiple, the second third). Then the end of the one-year buyback, about April 2027, with no promise of renewal ($0.0022, the rest). Regulation is a tail risk I do not price: a RICO claim against the company survived a motion to dismiss on August 31, 2026 (ref. 11), and UK users have been blocked since a regulator's warning (ref. 12).



Levels: sensitivity grid (Investment proposal 1) and trade plan. Dates: DefiLlama weekly and monthly prints, Blockworks vesting schedule, buyback policy (to Apr 28, 2027). The dashed line is a reading order, not a forecast of timing; any stop reading ends the path.

Fig. 5 — The path, dated: public events of the next six months and the grid's levels; grey band = the last two thirds of the entry on a rebound to $0.0055–0.0060 by November 16. Sources: DefiLlama, Blockworks filing, buyback policy.

Trade. A pair: short the PUMP perpetual and long SOL for the same amount, no leverage, 1 % of the book. The Binance perpetual trades $222M a day and open interest across venues is $352M (ref. 3), so even a $10M short is a small slice of it, and funding currently pays the short. Entry: a third at the October 21 price ($0.0047), and two thirds on a rebound to $0.0055–0.0060 before the November 16 print, or at market after that print if it confirms (the table uses $0.0053). With this much volatility, any price stop within reach is noise. So the stop is on the thesis, read on public series (weekly revenue, Allium's buyer count), plus a catastrophe stop if the PUMP/SOL ratio doubles on a daily close.

Six-month package per $ of notional	PUMP leg	Package, funding net of SOL leg	Probability
Bear · Base · Bull (PUMP –72 % · –54 % · +16 %)	+75 % · +58 % · –4 %	+76 % · +59 % · –3 %	25 % · 50 % · 25 %
Catastrophe stop (pair ratio ×2)	–100 %	–99 %	12 % under my drift, 38 % with none (rough)

Assumptions: entry $0.0053, SOL leg at the same notional (PUMP's beta to SOL has run at 0.9–1.2, so the pair is close to beta-neutral), funding +8 %/yr for three months. **This is a thesis trade. Weighted by my scenario odds it returns about +30 % of notional if revenue keeps drifting down as it did twice before (12 % chance of hitting the catastrophe stop), and about −8 % if the same scenarios face a stop as likely as for a ratio with no drift (38 %). Rough, and biased upward.**

Risk	Probability	Effect	Trigger	Response
The crowd returns and the multiple follows	medium	bull case $0.0055, or the stop	revenue ≥ $16M/wk for 4 weeks; buyers' 7-day mean > 155k/day (Allium) for 2 weeks	thesis stop
Take rate restored (it ticked up from 30 % to 37 % weekly since mid-September), buyback raised or topped up from the $1.32B raised (ref. 7)	low–medium	+20–40 %	weekly take rate above 40 %; config update	thesis stop; halve the position
Squeeze: OI 16 % of market cap, ~14 % of circulating supply on exchanges, ×3.6 this summer	medium	pair ratio ×2	listing, buyback news, SOL rally	catastrophe stop; 1 % of the book; no leverage
Funding turns against the short (seen Oct 17–19); product expansion (app, perpetuals) lifts revenue per buyer	medium	–15 to –25 % of notional; revenue holds while buyers fall	7-day funding below –20 %/yr; the monthly prints	halve the position; thesis stop, not crowd numbers

Conclusion

Short PUMP against SOL, to $0.0022, by April 2027. The price assumes October's crowd will stay. My data says it is leaving, and the buyback holding the price up has an end date. I played this game with my own money, and if my thesis stops trigger on the public series, I am wrong and I will say so.

References

Primary data (all pulled by the author; raw responses saved in the repository under `data/raw/`):

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15. Szwajcok, Tsuchiya, Liu, Soska, Payer, Christin, "Meme Coin Factories: Uncovering Large-Scale Manipulations on pump.fun", arXiv 2609.10246, ACM CCS 2026, https://arxiv.org/abs/2609.10246 — wash trading ≥ 17 % of trading transactions (strict lower bound), 1.32 % of volume; graduation 1.02 %.
16. Marino, Naviglio, Tarantelli, Lillo, "Predicting the success of new crypto-tokens: the Pump.fun case", arXiv 2602.14860 (Feb 2026), https://arxiv.org/abs/2602.14860 — graduation 0.63 %, median curve life 4.4 min.
17. Ding, Luo, Xu, "Decompose Market Manipulation Strategies: Evidence from On-chain Meme Coin Market", SBFC 2025, https://sbfc.sydney.edu.au/2025/papers/SBFC2025_4G3_P154.pdf — sniper and "rat" bots; creators as the most frequent dumpers.

18. Kamat, "Do Two On-Chain Observation Pipelines See the Same Tokens?", arXiv 2609.18975 (Sep 2026), https://arxiv.org/abs/2609.18975 — why collector coverage is reported.

Cross-checks (second pipelines):

19. Allium, Explorer (Academic plan), tables `solana.dex.trades` and `solana.assets.spl_token_balances_latest`; Source: Allium, https://allium.so. Public queries: q1 buyers per day, Oct 5–20 https://app.allium.so/s/OTno0JfpQFp077eJY5YTJg ; q1b programs carrying the curve swaps https://app.allium.so/s/XaiMvOaNZ7iDnTkM-DpzHA ; q2 daily activity on the Solana launchpads and PumpSwap, Feb 1 – Nov 3 https://app.allium.so/s/zliyTYXOn1IcA3NJ57Tz1w ; q3 concentration of PUMP holdings https://app.allium.so/s/uB82BHSqTK_TlLxwy1CNAw ; q4 crowd median, creations, graduations https://app.allium.so/s/QzEkzrtdR7w5SeVC612_HQ ; run Nov 4–6, 2026; queries committed before their first run (`crosscheck/` in ref. 5).
20. Blockworks Research API (competition access) — pump.fun project revenue, token-holder distributions (buybacks; no dividends), launchpad revenue by venue, PUMP supply and market cap; accessed Oct 31, 2026; https://app.blockworks.com.

Appendices (outside the 5-page count)

Reference material — definitions, the confidence table, the reconciliation of the money, the valuation arithmetic. The main report stands without it.

Glossary. *Hand / address*: one wallet (merging co-signing wallets changed nothing in this data); not a person. *Strict P&L*: profit or loss per position with tokens still unsold counted at zero. *Positive control*: a case whose answer is known in advance (a token that graduates must multiply ×14.7 on the curve); if the method finds it, the method works. *Second buyer*: share of new tokens that get a buyer from outside the creator's wallets within 60 seconds — a rough sign of real demand. *Take rate*: the share of trading fees pump.fun keeps as revenue. *Bonding curve / graduation*: the formula that prices a new token as people buy and sell; graduation moves the token to the PumpSwap exchange. *Perpetual, open interest, funding*: a futures contract with no expiry; the total outstanding; the periodic payment between longs and shorts. *FDV vs market cap*: all tokens that will ever exist, vs those in circulation. *Constant multiple*: assuming the price-to-revenue ratio stays where it is today.

Appendix A — The instrument

Collector. A read-only collector on 35 EVM chains plus Solana. It decodes launchpad events (bonding-curve buys and sells, pool creations, graduations, mints), stores them as hourly JSONL, and enriches each record with the real sender of the transaction (`tx.from` , not the router in the event), the deployer, and the size and simhash of the token's bytecode. Per-address P&L is reconstructed on bought tokens only (sales of tokens never bought — allocations, transfers — are excluded; this one rule removed several "winners" at absurd returns). Every test started with a written rule (thresholds, same-day controls, what would count as a positive result) and ended with a verdict against it; a rate-limited RPC returns "zero logs" without error, so every zero was challenged with an unfiltered canary request.

Known pitfalls, documented as they happened (each cost me a wrong conclusion): quoting a token at the last logged price instead of its historical on-chain state (a +198 % replay that was −18.9 % for real); a constant-liquidity simulator (off by ×30 on thin pools); the router counted as the buyer; dead pools left in a replay (17 % of them); an unmatched control; journals that count one side only — only on-chain balances are trusted for money.

Windows. pump.fun: 16 full UTC days (Oct 5–20, 2026, 384 h), ≈ 65.3M transactions, 395,253 launches; 23,030 of 23,040 minutes with a source and a canary at 99.4 % of sampled transactions (98.3 % worst day), which did not catch the loss of swaps after October 14 (cross-check below); one run per day, same parameters, every daily result published. Arc factory pools: 17 days (Oct 4–21). Robinhood Chain family: Oct 4–9. Other venues in Table B1: 6–22 h each (Oct 4–8).

Confidence — one line per proprietary number.

Number	n	Window (UTC)	Method	95 % CI	Test	p
Crowd median P&L, entry 20–60 s, non-graduated tokens: **−17.4 %** (valued at the last curve price −15.4 %; excluding the 11.2 % of positions never sold −12.2 %; closed positions only −11.6 %)	1,691,317 positions	Oct 5–20, 16 days	strict P&L on curve flows, curve fees included, unsold remainder = 0; each variant negative on 16/16 days (results/robustness_2060.csv)	[−17.5 %; −17.3 %] (order statistics, positions treated as independent); [−19.8 %; −16.0 %] = 95 % CI of the mean of the 16 daily medians (−17.9 %), by bootstrap	sign test, 16/16 days negative (days of one regime are not independent: indicative)	1.5×10^{-5}
Same, mean −24.3 %; losing positions 74.8 %	1,691,317	same	arithmetic mean; count	±0.1 %	—	—
Hour by hour: 384/384 hours negative (exact-curve replay, E = 60 s, H = 60 s)	384 hours	same	median per hour of birth	—	count only (hours of one regime are not independent)	—
Out of sample: replay cell chosen on the first half of each day, judged on the second	16 days	same	forward walk, split at mid-day	—	second half negative on 16/16 days (robust mean −6.9 % to −8.0 %; median −3.5 %)	—
Positive control: graduated tokens	9,916 graduated	same	theory $(115/30)^2 = 14.7$	—	—	—

Number	n	Window (UTC)	Method	95 % CI	Test	p
×14.70 birth → end of curve						
Creator sells ≤ 3 s after an outside buy: ×1.8–4.6 vs matched control (×2.1 once conditioned on prior third-party sales, 19-h sub-window)	2,780,579 outside buys	per day	matched control: same tokens, same age, no outside buy	—	ratio > 1 on 16/16 days	—
Active buyers −16 % week over week on Allium's count (163,333 → 137,677 per day); my collector −26 % (155,300 → 114,481), biased by its coverage loss after Oct 14	7 + 7 days	Oct 6–12 vs Oct 13–19	distinct addresses with ≥ 1 curve buy that day (Allium: signer of a swap that sells SOL)	—	same-weekday comparison, 7/7 pairs down in both pipelines; SOL −8 % is context, not a control	—
Factory pools tracked on Arc: 93 % of trading addresses end negative	7,160 addresses, 1.54M trades	Oct 4–21, 408 h	realised-or-written-off (open positions at last curve price, zero if the pool was drained)	92.9 % ± 0.6 pt (binomial)	no regular winner (≥ $20/day in two of three 136-h windows)	—

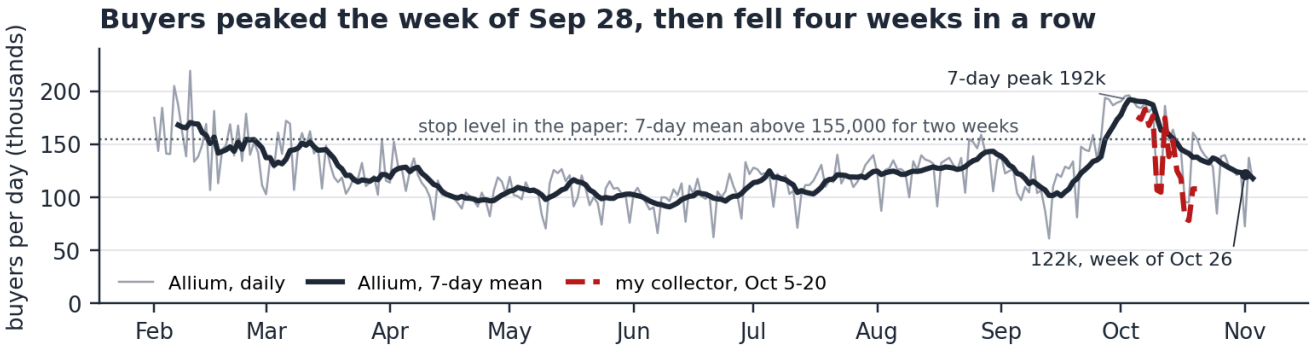
Cross-check with a second pipeline (Allium, ref. 19). Definitions and tolerances were written on October 26 and published in the repository on November 1, before the first Allium query on November 4; each query was committed before it was run (`crosscheck/` in the repository). Most tolerances fail. The buyer and LaunchLab gaps come mainly from my collector, which loses coverage after October 14; Allium's buyers also include the mayhem launch mode, which my count leaves out. The creation, graduation and crowd-median gaps do not come from that loss: my collector counts more creations and graduations than Allium, and its daily medians already average −17 % on October 5–14. I do not explain those.

measure, Oct 5–20	my collector	Allium	tolerance	verdict
buyers per day, Oct 6–12	155,300	163,333 (+5 %)	±10 %	pass
buyers per day, Oct 13–19	114,481	137,677 (+20 %)	±10 %	fail
week on week	–26 %	–16 %	within 5 points	fail (same sign, 7/7 pairs down in both)
crowd median, 20–60 s	–17.4 %	–13.2 % (2.25M positions)	±2 points	fail (same sign)
creations, all launch modes	700,261	662,977 (–5 %)	±5 %	fail, at the edge
graduation rate, all modes	2.08 %	1.67 %	±0.3 point	fail
LaunchLab swaps per day, week on week	–71 %	–54 %	±10 % on levels	fail (same sign)
distinct buyers, 16 days	1,129,662	1,296,189 (+15 %)	±10 %	fail (coverage loss; Allium also counts the mayhem mode)
creators' P&L	–28.8 %	+28.9 %	±5 points	fail (Allium's creator is the first swap's signer, often a sniper)
second buyer within 60 s	51.3 %	69.2 %	reported, not calibrated	—

The crowd in three earlier fortnights (Allium, run on November 9). The same crowd query, written and published on November 5 for February 1–16, May 1–16 and August 1–16 (`q4b` , `q4c` , `q4d`), was meant to run only if the October query fell within tolerance. It did not, so this run is a departure from my rule 2, made at my request on November 9 and stated here; the numbers are Allium's, second pipeline, not calibrated. Crowd median, buyers 20–60 s after launch, strict P&L: **−15.0 %** in February (1,802,791 positions), **−12.2 %** in May (1,571,731), **−13.6 %** in August (2,057,007), against −13.2 % in October (2,249,019). Graduation rate 0.8 %, 0.7 % and 2.2 %; second buyer within 60 s 65 %, 70 % and 69 %. Queries, results and run log are in the repository (`crosscheck/`).

Day by day, Allium counts 3–11 % more buyers than my collector until October 14, then 19–22 % more on October 15–18 and 44–49 % more on October 19–20. Allium's swap count on the curve stays flat (3.9–5.1 million a day); my collector, counting all launch modes, saw 89 % of it on October 5, 82–91 % until October 14 and 52 % on October 19. Every curve swap goes through the same pump.fun program, so this is a loss on my side, not a new program; its

cause is not established, and my canary, which samples transactions, did not catch it. Allium's creator proxy (the first swap's signer) is too coarse for the creators' P&L, so that row fails and I do not use Allium's value. My rule 2 said that a second pipeline which fails calibration is shown but not used as evidence. For the buyer counts the failure is mainly on my side, so there I depart from that rule, and say so here; the departure works against my thesis, because Allium's fall (−16 %) is smaller than my collector's (−26 %); for the crowd median the body keeps my number and prints Allium's beside it. Consequences in the main text: the weekly fall and the buyer stop use Allium's count; the crowd median keeps my number, with Allium's beside it; the food chain (Fig. 3) and the per-position results are my collector's, and their second week is undercounted.



Source: Allium (allium.so), solana.dex.trades, protocol 'pumpfun'; a buyer is the signer of a swap that sells SOL. Query q2 in crosscheck/queries/allium/. Collector: docs/pumpfun/03 §10. The two agree within 3-11 % to Oct 14, then the collector loses coverage (crosscheck/results/allium/README.md).

Fig. A1 — Active buyers on the pump.fun curve, February 1 – November 3, 2026, Allium's count (daily and 7-day mean), with my collector's sixteen days. Second pipeline, not calibrated (above). Source: Allium (allium.so), query q2 (ref. 19).

Caveats: the 16 days are one market regime (post-peak October 2026) and consecutive days are not independent, so the day-level sign test is indicative, not a proof of generality; the daily medians (−12 % to −26 %) are the honest spread. Kamat (2026, ref. 18) shows that two collectors can see almost disjoint token sets, which is why coverage is reported. Addresses are wallets; the food-chain script's co-signer grouping returned the same count (1,129,662), so no merge is applied.

What my method misses. Sales after graduation on PumpSwap are invisible, so P&L on graduated tokens is a lower bound. The strict rule (unsold tokens at zero) overstates the loss on tokens still alive at the end of the window: valued at the last curve price the median is −15.4 %, and −12.2 % excluding never-sold positions; the last two days carry twice the usual share of never-sold positions (18–21 % against 5–15 %), which drives their strict medians to −25 % while the rest of their positions sit at −13 % to −15 %, in line with the other days; the observation horizon is the same every day (each day is followed to 00:00 UTC the next), so this is not censoring; but these are also the two days my collector saw least of the flow (cross-check above), so missed sales may explain part of it; on October 5–14, the better-covered days, the daily medians average −17 %; across days the never-sold share tracks the strict median ($r = -0.89$) but not the median of the rest ($r = -0.5$). Bots and wash trades are not filtered out of the crowd; if anything they make its median *less* negative. An address is not a person. Solana transaction, priority and trading-bot fees are not deducted, so the crowd's true P&L is worse than shown; on EVM chains gas is not deducted and 4.7 % of Arc senders are unattributed. My graduation rate (2.5 %) is above the literature's 0.6–1.0 % (refs 15–16); a mania month and a different perimeter explain part of the gap, I do not fully explain it, and I cite both.

Appendix B — Five launchpads under the same instrument

Table B1 — Five venues under the same scanner. The busiest launchpad still costs the crowd 17 % per position. Read signatures, not levels: the other venues are snapshots of a few hours to a day (Oct 4–8). Source: my collector; columns below.

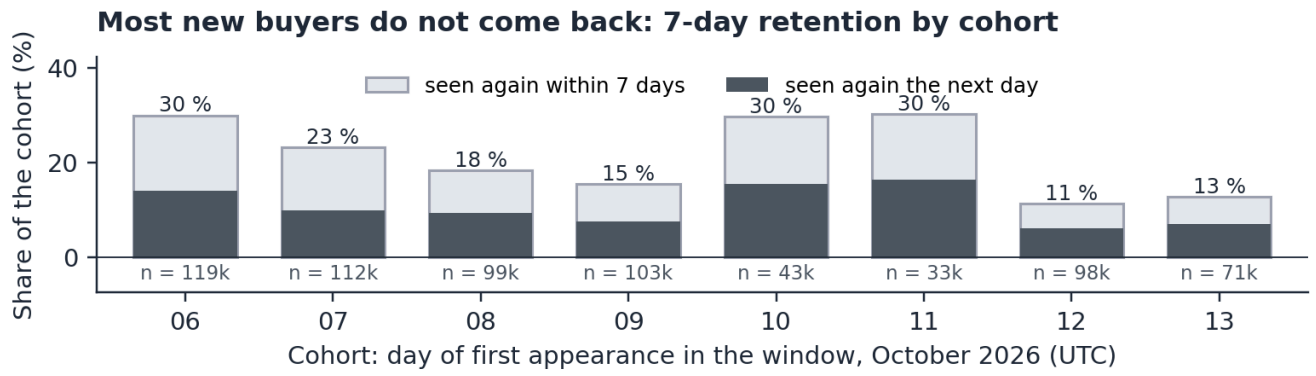
Venue (chain)	Launches per hour	Second buyer ≤ 60 s	Crowd median P&L	Operator pull or sell within seconds of an outside buy (× matched control)	Inorganic signature
pump.fun (Solana), 16 days	**1,029**	**51 %**	**–17.4 %**	**×1.8–4.6 (×2.1 controlled)**	**12 % never trade; buyers –16 % w/w (Allium)**
PumpSwap (post-graduation, 19 h)	36–51 playable pools	100 %	–20.5 % night / +1.7 % day	none	virtual reserve counted in price
LaunchLab (Raydium, other quote tokens)	77	93 %	–23.2 %	×10.4	StonkFun = 95 % of September launches
Zora	138	56 %	–0.6 to –2 %	none (0 / 606)	one operator = 83 % of births
Pons V2	354	n/a	–5.9 % (3rd-party)	5 of 6 drained in 1.4 s	fees –43 % after gas subsidy ended

Columns of Table B1. *Launches/h*: new tokens or pools per hour. *Second buyer*: share of new tokens with at least one buy within 60 s from an address outside the creator's cluster — my proxy for organic demand. *Crowd median P&L*: for pump.fun, the strict P&L of entries 20–60 s after launch; for the other venues, third-party buys within 600 s of birth (gas not deducted). *Operator pull or sell after a third-party buy*: how much more often the operator pulls liquidity (EVM venues) or the creator sells (pump.fun) within seconds of an outside buy than at matched moments on the same tokens at the same age (pump.fun ×1.8–4.6 by day; ×2.1 once conditioned on prior third-party sales, 19-hour sub-window — the raw ratio overstates targeting). *Inorganic signature*: the clearest sign of manufactured activity on that venue. P&L rules differ by venue — pump.fun strict (unsold remainder = 0), Arc realised-or-written-off (open positions at the last curve price, zero if drained) — so the table compares signatures, not levels.

Factory pools I tracked on Arc, 17 days (Oct 4–21). 7,160 trading addresses; 508 end positive (+$38.1k) and 6,652 negative (−$46.1M); 1.54M trades; no regular winner (≥ $20/day in two of three 136-hour windows); median crowd position $146 (118-h sub-window, Oct 4–9). The 7,160 exclude 11,680 liquidity addresses (deployers and drains); the dollar sum (mean loss $6,930 against a median of $47) is probably mostly the factories' own capital cycling through their pools, which is why I cite the count and the median, not the sum.

Same-chain control. LaunchLab (Raydium), the other Solana launchpad the collector listens to: swaps per day 122,640 (Oct 6–12) → 35,484 (Oct 13–19), −71 %, in my collector (COMPUTED from the daily `verificationsLaunchLab.echanges`); Allium's count is 172,442 → 78,660, −54 %, back to 165,524 a day in the week of October 26 (q2), against −3 % for pump.fun's curve swaps and −16 % for its active buyers. Both of my collector's week-two numbers are undercounted (Appendix A).

pump.fun, 16 days (Oct 5–20). 1,029 launches/h for SOL-quoted tokens (1,506/h on Oct 5 → 705/h on Oct 20, a fall that my coverage loss and the shift to other launch modes exaggerate; all launch modes 1,824/h, 700,261 tokens, 2.08 % graduated; the SOL-quoted family is 59.6 % of curve swaps); 51.3 % second buyer (44.6 % strict); exact-curve replay at E = 60 s, H = 60 s: median −2.5 % (the round-trip fee) on each of the 16 days; at the replay cells closest to the recipe (entries 20–45 s, exits 120–240 s later; the bots held 110–240 s) the median is −2.7 % to −7.3 %, negative 16/16 days in each cell; after-graduation entries on PumpSwap (5–45 s, 19-h sub-window only): median −20.5 % at night, +1.7 % by day. Users (my collector, week two undercounted, Appendix A): active buying addresses 155,300/day (Oct 6–12) → 114,481/day (Oct 13–19), −26 % (Allium: −16 %) while SOL fell 8 %; 7-day retention of a day's new entrants 11–23 % for the weekday cohorts from Oct 7 (next day 6–10 %), ~30 % for the Oct 6 cohort (it still contains regulars who merely skipped the first day) and for the two small weekend cohorts (Fig. B1); revenue per active buyer-day $17 on my count, about $15 on Allium's; fees per active buyer-day averaging $47 ($36–65 by day), inflated after October 14 by the undercount.



Source: author's Solana collector, pump.fun/SOL, cohorts of buyers not seen earlier in the window (Oct 6–13; the Oct 5 cohort is excluded because on day 1 every buyer is 'new'); n = cohort size.

Fig. B1 — Seven-day retention of each day's new entrants on pump.fun (cohorts Oct 6–13; the Oct 6 cohort is inflated by regulars absent on day 1; the two weekend cohorts, Oct 10–11, are small and more loyal; follow-up after October 14 undercounted, Appendix A, so the later cohorts' retention is biased low). Source: my Solana collector.

The food chain by hand (Fig. 3). Hands grouped by the number of positions taken in the 16 days, non-graduated tokens, strict P&L. The all-tokens variant (graduated included) gives −70 % for single-position hands and −3.7 % above 2,000 positions; it is biased against graduated tokens' buyers, whose sales happen on PumpSwap outside this data, so the body uses the non-graduated variant, consistent with Table B1 and Fig. 1.

Hands by positions in 16 days	Hands	% of hands	% of positions	% of SOL staked	Median strict P&L	Stake-weighted	% losing
1	507,199	44.9 %	3.2 %	3.2 %	−46.4 %	−19.8 %	84 %
2–5	437,462	38.7 %	8.7 %	6.0 %	−59.6 %	−24.1 %	84 %
6–20	115,276	10.2 %	8.3 %	6.8 %	−26.7 %	−18.7 %	79 %
21–100	53,404	4.7 %	17.1 %	15.1 %	−18.3 %	−16.0 %	75 %
101–500 (≈ 6–31 a day)	14,474	1.3 %	21.3 %	21.8 %	−13.4 %	−11.8 %	73 %
501–2,000 (≈ 31–125 a day)	1,452	0.1 %	9.7 %	9.9 %	−6.9 %	−7.9 %	69 %
> 2,000 (≥ 125 a day)	395	0.03 %	16.8 %	19.3 %	−4.0 %	−11.1 %	67 %

The "% of positions" and "% of SOL" columns are relative to the all-token totals (they sum to 85 % and 82 %); relative to the non-graduated totals, hands with five positions or fewer hold 14.0 % of positions and 11.2 % of the SOL, hands above 100 positions 56.2 % and 62.2 %. Script profile (more than 100 positions and active on at least 10 of the 16 days, all tokens; creators and snipers as scripts: Ding et al., ref. 17): 10,819 hands, 0.96 % of hands, 40.9 % of positions, 49.5 % of SOL staked. Occasional profile (five positions or fewer, all tokens): 944,661 hands, 83.6 % of hands, 15.4 % of positions, 13.4 % of SOL.

Holder map of PUMP (Solscan, October 22, 2026; top 80 accounts read with labels). 386,417 holders on Solscan's count (Allium, counting owner addresses with a positive balance: 423,446 on November 5 and 431,676 on November 6, ref. 19); top 10 = 60.3 % of the supply left after burns (Allium, November 5: 59.4 %), top 100 = 88.6 %. By category: pump.fun custodian and allocation vaults 329.9B (39.9 % of the 826.8B left after burns, ≈ the non-circulating supply); round-number Squads multisigs (19 accounts), 196.6B (42.6 % of circulating), several in identical tranches (3.125B ×3, 1.562B ×2); the same tranches also appear among unlabelled wallets (2.087B ×4, 3.125B ×1, 1.562B ×3) — read as unlocked insider or private-sale allocations held (at most 103B had unlocked to insiders by October 22), an inference from the amounts, not a Solscan label; exchanges and the Hyperliquid bridge 65.6B (14.2 %: Binance 27.4B, Kraken 11.1B, Hyperunit 9.0B, OKX 8.4B, Bybit 4.8B); unlabelled large wallets 118.5B (25.6 %); ranks 81 and beyond hold 113.5B (13.7 % of total supply). About 35B of non-circulating supply sits outside pump.fun's labelled vaults, so the multisig and whale shares of circulating supply above are upper bounds. Raw table: `data/raw/solscan_pump_holders_2026-10-22.md` .

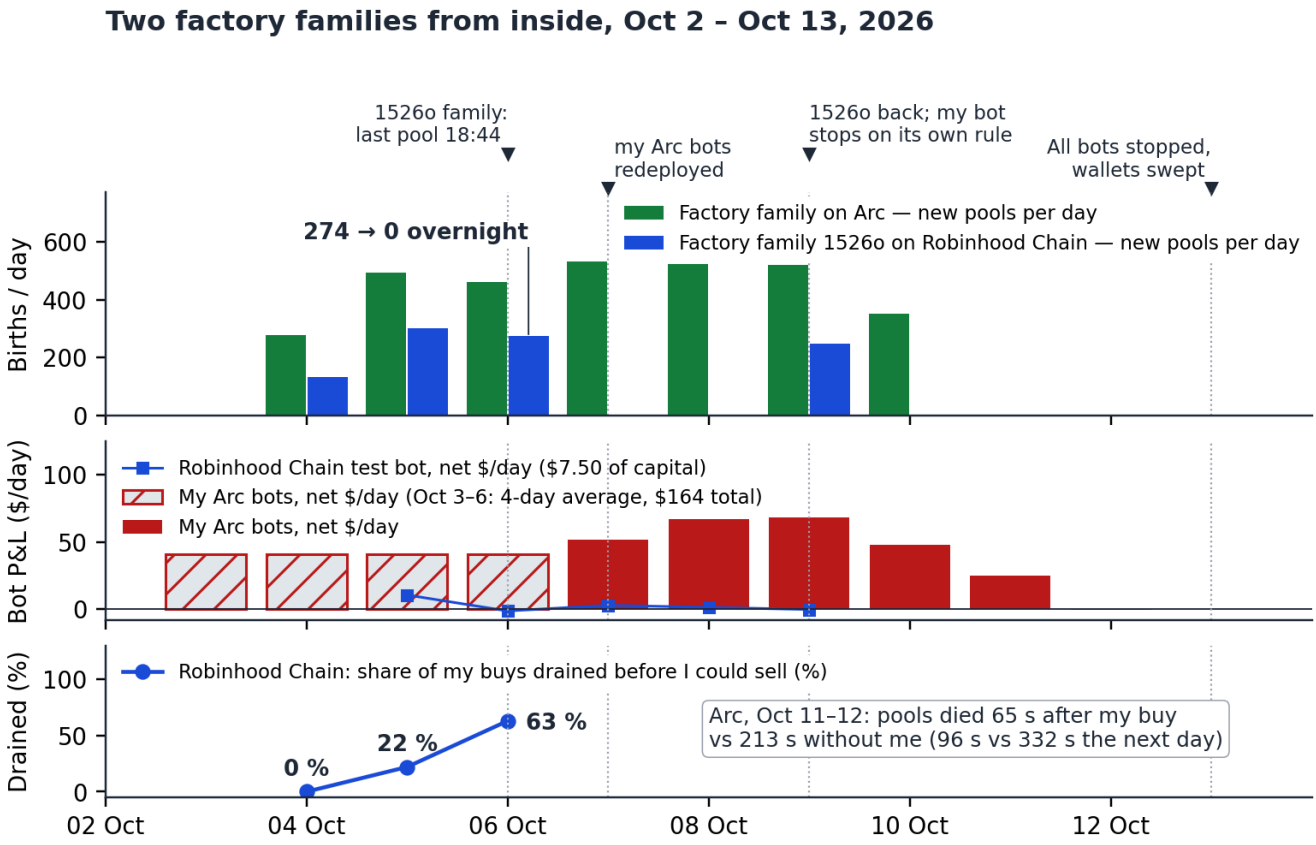
Appendix C — The hunt, day by day

Reconciliation of the money (on-chain where possible).

Item	Amount	Basis
External capital in, floor	$55.66 USDC on Arc, on or before Sep 23 (origin of this first deposit not reconstructed)	bot ledger

Item	Amount	Basis
External capital in, ceiling	+ $10 (Sep 24) + $19.36 (Sep 25–29) + $0.90 (Sep 30) + $33.61 (Robinhood Chain, by Oct 2; bridged from Arc or from the vault, undetermined) + $2.93 (Base) = **$122.46**	lab notes
Balance trajectory (native balances, open positions excluded)	Sep 25 $40.81 → Oct 1 $201.24 → Oct 3 $330.29 → Oct 5 $510.42 → Oct 7 $579.50 → Oct 13 ≈ $804	balance snapshots
Recovered on Oct 13	640.77 USDC at the vault (four sweeps below = 629.14 USDC, plus 11.63 USDC from reserve wallets) + 0.042072 ETH (≈ $113 at $2,695/ETH) + 50 USDC converted to 0.176 TAO on Oct 8 (≈ $50) = **≈ $804**	on-chain
Campaign net, Sep 23 → Oct 13, four chains	$804 − $56 to $122 = **+$682 to +$748**	computed
Journals (one side only), Oct 3–12, Arc	≈ **+$424**; the balances over the same days went $330 → $804 (+$474): journals under-count by ≈ 10 %, which is why money is read on balances, not journals	journals vs balances
By chain	not separable exactly: the bridges Arc → Robinhood Chain (Relay) and Arc → BSC (Across) are internal; Arc carried ≈ 90 % of the P&L	—
September tests on Pons (Robinhood Chain)	≈ −0.26 ETH ≈ −$700 at $2,695/ETH (the September ETH price is not documented)	expense ledger
Overall, early September → Oct 13	**≈ −$20 to +$50 — about zero**	computed

53 addresses on Arc, Robinhood Chain, Base and BSC; the bots' parameters were unchanged between Oct 4 and 6 (journal). The "~$800" sometimes quoted is the final balance, not a gain.



Source: author's bot journals and lab reports, reconciled to on-chain balances (Appendix C). Days without a reported value are blank.

Fig. C1 — Two factory families from inside (Arc and Robinhood Chain): births per day (not logged after Oct 10), my bots' net dollars per day (journals, one side only; the Oct 3–12 entries sum to ≈ $424 of the $682–748 campaign), the share of my buys drained. Source: my collector and bot journals, Oct 2 – Oct 13, 2026.

Public proofs. The vault also served as the funding wallet of the campaign (its balance before the campaign is not part of the P&L). Sweep transactions on Arc (chain id 5042) to the vault 0x7d85bF7a82470837A1d832e4fa503a7ebF20ca97:

- Oct 7, 2026 19:23 UTC — w000 → vault, 196.739407 USDC: `0x4278048243806f9effb75206445a33b6c29fb4519224220df66470b4b84c163a`
- Oct 7, 2026 19:23 UTC — w001 → vault, 178.116421 USDC: `0x4c6db445c8349f39b5b3e40b4ea30339b2127f1a7d7bb4ce2b601abc0aef73ed`
- Oct 13, 2026 21:56 UTC — f101 → vault, 143.872356 USDC: `0x99e5f0e954924dba3378e954fb8e67b2596e99d0de2edfe99093d4ce8812917e`
- Oct 13, 2026 21:56 UTC — f102 → vault, 110.408233 USDC: `0x8dd41acced7790ea74812702f9fc50cd1ff9e6d6b149bd66f692074f1e27cced`

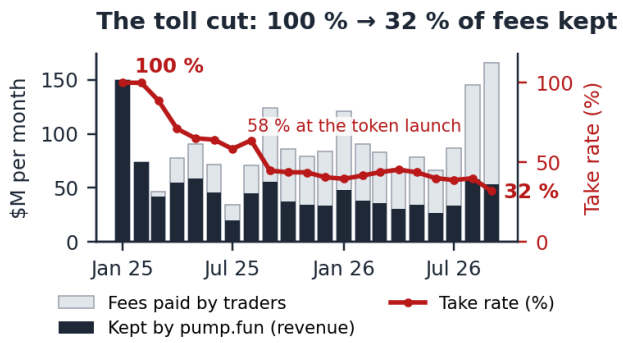
Addresses f101 and f102 can be found by a third party among the winning addresses of Arc's factory pools in the Oct 4–9 window from public logs alone (the Oct 9 lab report; over the full 17 days no address meets the "regular winner" rule of Appendix A). The day-by-day table (births per family, net dollars, pool life with and without my buy, share of my buys drained, events) is published with the data (`data/bots-daily.csv`).

Appendix D — Valuation arithmetic

Target. Price = multiple × annual revenue ÷ circulating supply (461.9B). Multiple today = market cap $2.19B ÷ annualised trailing-30-day revenue $836M = 2.62×. Pre-mania base = Feb–Jun 2026 average $33.2M/month = $398M/yr. 2.62 × $398M = $1.04B ÷ 461.9B = $0.00226; at the 2.6× quoted in the text it is $0.00224, so **$0.0022 (−54 %)**. The multiple has been pro-cyclical (1.9× at the June 2026 trough, 2.6× at the peak), so a constant multiple is the conservative assumption; the grid on page 4 shows the alternatives.

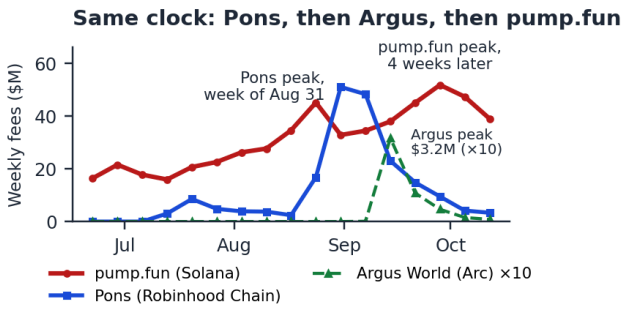
Reverse check (fully diluted). FDV $3.92B ÷ annual holder flow (50 % × $836M = $418M) = 9.4×. At a 25 % discount rate, r − g = 10.7 % → implied perpetual growth g ≈ 14 %/yr; the revenue that justifies today's FDV with no growth is R* = $3.92B × 0.25 ÷ 0.5 = $1.96B a year, forever ($1.10B on circulating market cap), against $836M, the last 30 days annualised. Flat peak holder flows ($418M a year) are worth $418M ÷ 0.25 = FDV $1.67B (−57 %); base flows ($199M) $0.80B (−80 %).

Buyback-coverage price (no multiple). Net supply is zero when 50 % of annual revenue buys back exactly the annual insider issuance (82.5B tokens): P = 0.5 × revenue ÷ 82.5B. Run-rate $836M → $0.0051 (today's price is within 7 % of it); Feb–Jun base $398M → $0.0024; June trough $319M → $0.0019.



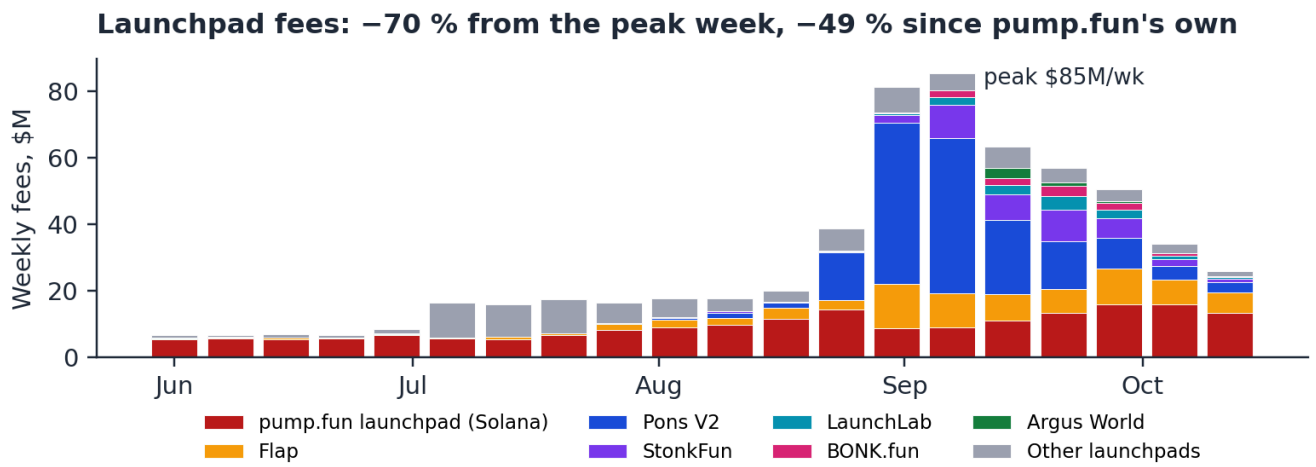
Source: DefiLlama dailyFees vs dailyRevenue, pump.fun parent, Jan 2025 – Sep 2026. Take rate = revenue / fees.

Fig. D1 — The take rate: 100 % → 58 % → 32 %.



Source: DefiLlama dailyFees, weeks of Jun 22 – Oct 12, 2026 (week of Oct 19 partial, excluded). Argus World scaled ×10 for visibility.

Fig. D2 —Three launchpads, one summer: weekly fees peaked in launch order. Source: DefiLlama.



Source: DefiLlama dailyFees, 16 launchpads with >$0.3M fees/30d in October (all chains), pump.fun counted as its launchpad only; ISO weeks 23–42 of 2026. Universe fixed on October fees (earlier casualties absent).

Fig. D3 — Launchpad fees, launchpads only (16 venues, all chains; for pump.fun its launchpad, not PumpSwap, the Terminal or the app); −70 % from the pie's peak week (Sep 7), −49 % since pump.fun's (Sep 28). Pons is the largest slice of the peak week (57 %); universe fixed on October fees. Source: DefiLlama. Shares depend on how DefiLlama splits pump.fun's products, so I read this figure for the size of the pie only: on revenue across 43 venues pump.fun never fell below half (50.6 % at its lowest, week of September 7; Blockworks, ref. 20).

What the 2025 long said, and what I measure.

They said (the 2025 finalist long; the bull case today)	I measure	Where
85 % share of launches is a moat	the moat held: never below half of launchpad revenue (Blockworks) and 70–94 % of Solana launchpad swaps (Allium); but the launchpad pie is −70 % from its peak week	Fig. D3
~100 % of revenue to buybacks; "$490M bought back"	50 %, by decision; 18¢ per fee dollar vs 58¢ at launch; unlocks outweigh buybacks at base revenue	Fig. D1
$870M of annualised fees, growing; target $0.0152	−87 % and −52 % from both prior peaks, −22 % from this one; over the last year the market paid 1.6× to 4.2× trailing revenue (month-ends), 1.9× at the June trough, 2.6× today	Fig. 4

Appendix E — Reproducibility

Every number recomputable from public data or from my published collector outputs has a command in the repository; the bot numbers (Fig. 2, Appendix C) are typed from my journals and reconciled to on-chain balances (https://github.com/JeanBaptisteDurand/UBC_Research_Competition, ref. 5; numbers marked C in its README need the per-position files (3.9 GB) or the hourly collector files (8.6 GB), available on request): the raw API responses (`data/raw/` : DefiLlama, CoinGecko, Binance, Hyperliquid, the Solscan holder table), the 16 daily pump.fun results and their aggregates (`data/pumpfun-16d/`), the Arc 17-day ranking (`data/arc-17d/`), the daily bot table (`data/bots-daily.csv`), the lab tools (`scripts/lab/` , JavaScript, read-only, published; they run once per UTC day on the collector's hourly files, 8.6 GB compressed, available on request, as are the 3.9 GB of per-position files) and the figure scripts (`scripts/` , Python 3, matplotlib). Cross-checks: Blockworks (ref. 20) agrees within 1 % with my sources on base revenue (DefiLlama), cumulative buybacks (pump.fun's page) and circulating supply (CoinGecko); it shows half of net revenue going to buybacks since April 28 (the policy is kept) and no dividend. Two Blockworks counts do not fit mine: 779,657 launches on October 5–20 against my 700,261 (+11 %, outside my ±5 % tolerance), and curve swaps only 3 % lower week on week, as in Allium, while buyers fell 16 %. The Allium cross-check (ref. 19) is in Appendix A, with its failures; queries, results and verdicts are in `crosscheck/` , and the queries are public in Allium.